DOCUMENT // KNO-BRAND-001 // REV 2.0

KNOWINGSLAB

Brand & Content Strategy

Prepared: December 2025 | Revised: April 2026 | Author: Tom Leh

Channel Identity

Core Statement

Knowingslab is a robotics and space-focused channel following an engineer who builds things that work—actuators, arms, rovers, and occasional detours into other domains when a problem catches attention. The content is technically legitimate (with companion blog posts for depth) but presented with personality and dry humour; the goal is making real engineering accessible without dumbing it down.

This is not a presenter explaining other people's work—it's documenting personal projects, including failures, late-night debugging, and the parts where months of progress get thrown out to start again.

Naming Convention

Element	Usage
Knowingslab	Primary channel name (compound, one word), spoken identity, handle (@knowingslab)
[K•]	Primary mark — HUD-style K with kinematic pivot at vertex (see §2 Mark). Renders as `ks-mark-hud` from `knowingslab-brand.typ`.
tomleh.gg	Portfolio/blog domain, companion technical content

Project Naming Convention

Projects use twisted acronyms that are descriptive but playful. This creates memorable names that reward attention.

Name	Meaning
ALGERNON	Articulated Limb, Generally Earnest, Rarely Nimble, Often Nuisance
NEIL Rover	Non-Expensive Intelligent Loadout Rover
DOTT	Don't Over Think That (LLM assistant concept)

Positioning

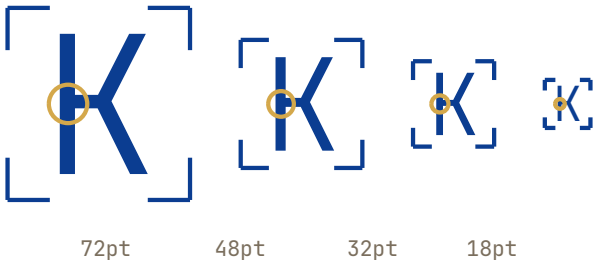
Reference Points	Differentiation
Joe Barnard (BPS Space)	Dry British tone vs American earnestness
Michael Reeves	More technical depth, less chaos-for-chaos-sake

Stuff Made Here	Scrappier, personal projects vs polished builds
Tom Scott	Similar articulate delivery, but maker-focused

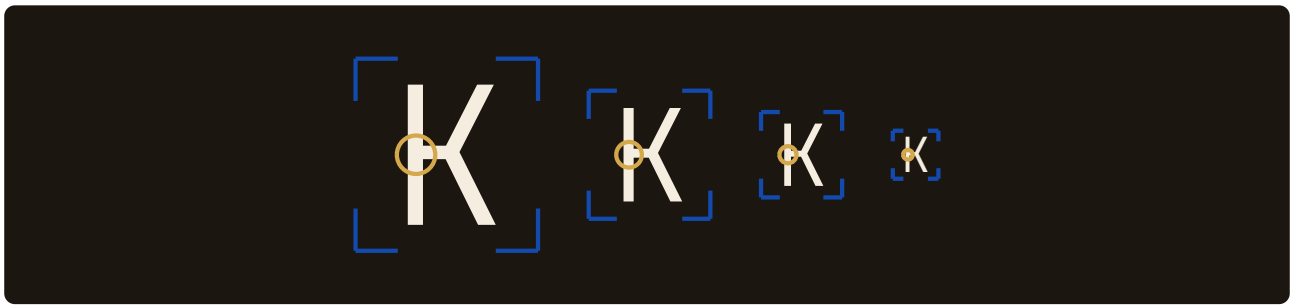
Visual Identity

Mark

The Knowingslab mark is a bracketed K with a pivot circle at the vertex — dual-reading as the letterform and as the kinematic joint symbol used in robotics schematics. HUD-style corner brackets frame the letter; everything renders at consistent thin stroke weight for the wireframe / targeting-reticle feel drawn from Elite Dangerous HUD language and Fraser's Project Hail Mary cinematography.



On the warm-dark background (primary context — thumbnails, video overlays, end cards), the mark inverts: letter and frame become tungsten cream, pivot stays gold.



Specification

Parameter	Value
Form	Corner-bracket frame + K + outline pivot circle at vertex
Line weight	1.6pt (default); scales with mark size
Corner length	0.22em relative to mark size
Pivot radius	0.10em
Pivot offset	-0.16em horizontal from K centre
K weight	Light (JetBrains Mono)
Letter + frame	NASA Blue on light; Tungsten Cream on dark
Pivot	Thermal Gold in both contexts

Usage

- **Avatar / favicon:** mark alone, centred, minimum render size 24pt (32pt+ preferred for mobile)
- **Watermark:** mark at 5% of frame height, bottom-right of video frame
- **End card:** mark + wordmark lockup (`ks-lockup`)
- **Inline / in-document:** use module tokens, never hex literals or redrawn SVGs

Wordmark

The wordmark **Knowingslab** renders in JetBrains Mono Light to match the K's stroke weight. Used for the channel banner, end cards, document headers, and any context where the name needs to be spelled out rather than glyph'd.

Knowingslab

Bare wordmark — preferred for banners and long-form contexts.

[Knowingslab]

Bracketed wordmark — use when pairing with the HUD mark system.

Canonical source

All mark variants (`ks-mark-hud`, `ks-wordmark`, `ks-watermark`, `ks-lockup`) are defined in `knowingslab-brand.typ`. Import and call those functions rather than redrawing by hand — tuning the mark there updates everything downstream in a single edit.

Colour Palette

Palette aligned to Project Hail Mary interior cinematography (Greig Fraser, 2026): warm analog grade rather than cold clinical. NASA Blue primary paired with Thermal Blanket Gold accent (the colour of MLI spacecraft insulation — JWST, Perseverance descent stage, Cassini). Tertiary accents cover the cool-morning and warning registers of Fraser's circadian lighting system.

Accents

Light/dark rule for primary: NASA Blue `#0B3D91` is canonical on light grounds (documents, print, white backgrounds). On warm-dark grounds the primary crushes into the background, so use the **dark-mode variant** `#124AAD` (slight lift, same NASA character) for frames, HUD elements, and wordmarks rendered on dark. The token is `ks-primary-bright`.

NASA Blue — Primary (light)	NASA Blue — Bright (dark)	Thermal Gold — Accent	Instrument Teal — Cool
#0B3D91	#124AAD	#D4A84B	#4A8287

Safety Vermillion — Warn
#C44A2F

Neutrals

Warm Black — Background	Tungsten Cream — Text	Warm Grey — Muted	Warm Rule — Divider
#1C1611	#F5EDDF	#7A6E5D	#B8AE9D

Canonical values live in `knowingslab-brand.typ` as tokens (`ks-primary`, `ks-accent`, `ks-accent-cool`, `ks-accent-warn`, `ks-bg-dark`, `ks-text`, `ks-muted`, `ks-rule`). Import and use tokens rather than hex literals in downstream docs.

Typography

Single-family hierarchy — JetBrains Mono carries everything (mark, body, HUD overlays, thumbnail titles). No display sans. Hierarchy is weight-based, not family-based.

- **Primary**: JetBrains Mono (required)
- **Fallbacks**: Space Mono, IBM Plex Mono

Weight tokens

Use these semantically rather than picking weights ad hoc. Tokens defined in `knowingslab-brand.typ`:

Token	Weight	Use
`ks-w-decorative`	ExtraLight	Subtle overlays, quiet marks
`ks-w-mark`	Light	The K inside the mark; wordmark
`ks-w-body`	Regular	Default body text
`ks-w-emphasis`	Medium	Inline emphasis, small headings
`ks-w-heading`	Bold	Section headings
`ks-w-hero`	ExtraBold	Thumbnail hero titles

Rationale

Monospace throughout gives the industrial/technical aesthetic anchor (mission control, spacecraft consoles, JetBrains Mono's code-editor origins). Single-family feel is critical for the HUD/reticle system to read coherently; a sans display face would fragment that. If a specific thumbnail needs more impact, reach for ExtraBold, not another family.

Visual Language

Analog Space-Industrial

(Formerly "NASA-Punk / Space Industrial" in v1.4; renamed to capture the warm/analog dimension central to the PHM-aligned direction.)

- Clean, functional aesthetic inspired by **lived-in spacecraft interiors and mission control** — Hail Mary, Nostromo, Solaris, 2001 Discovery. Industrial but warm, not clinical.
- **Warm-dark** backgrounds with tungsten-cream highlights; reject cold clinical whites and pure blacks.
- **HUD / reticle framing** — corner brackets, thin consistent line weights, targeting elements. The mark is the primary exemplar of this language.
- Grid-based layouts reminiscent of engineering schematics and spacecraft console UIs.
- Subtle space-hardware references: MLI thermal blankets, anodized aluminum, illuminated panels, circadian lighting cycles.

- **Optional textures:** rainbow-flare highlights (per Fraser), CRT scanlines, analog film grain, practical-light bokeh.
- Monospace typography throughout — single family, weight-based hierarchy.

Thumbnail Style
- Warm-dark background (#1C1611); reject cold black
- Corner-bracket framing — reticle motif carried over from the mark
- Hero title ≤6 words in ExtraBold tungsten cream; optional subtitle in gold, Regular or Medium weight
- HUD mark watermark bottom-right, 5% of frame height
- Clear primary subject: hardware, failure, result, or face
- **Face policy:** allowed when the video is personal (ALGERNON intro, career milestones, reflections). No face-performance tropes — no pointing, no shocked-face, no mouth-agape reactions. Composed, considered.
- **Accent usage:** gold for highlights and labels, teal for cold/technical moments, vermillion for warnings and pivot-points (sparingly)

Tone & Delivery

Voice Characteristics
- **Dry British humour** — understated, self-deprecating, deadpan
- **Technically precise** — correct terminology, no dumbing down
- **Conversational authority** — knows the subject, doesn't lecture
- **Slightly "posh"** — natural vocabulary elevated, not performed

What This Sounds Like

> *"So I've spent the last four months designing an actuator around cycloidal reducers. Quite pleased with myself, really. Then I saw a tweet about a Korean paper and immediately threw all of it in the bin. Let me explain why that was actually the correct decision."*

What to Avoid
- American "maker energy" enthusiasm
- Explaining obvious things to pad runtime
- Excessive self-deprecation that undermines credibility
- Winking at the "British" delivery (let it be natural)
- Apologies for technical content

ALGERNON

Overview
ALGERNON — Articulated Limb, Generally Earnest, Rarely Nimble, Often Nuisance

ALGERNON is a 5-DoF robotic arm that serves as the channel's recurring character and workshop companion. Named after the mouse in *Flowers for Algernon*, the reference implies something about potential, limitation, and affection despite inadequacy—without being heavy-handed about it.

Character
ALGERNON is **not** stupid. He's cat-like: selectively competent, curious to the point of mischief, and smarter than he lets on. The comedy comes from agency, not incompetence.

Personality Traits

- **Curious** — investigates things he shouldn't, reaches into drawers, examines components
- **Selectively helpful** — capable of handing you the right tool, but might not
- **Mischievous** — not malicious, but definitely chaotic neutral
- **Expressive** — uses wrist pitch/roll to "look" at things, tilt quizzically, recoil

What ALGERNON Does

- Appears in frame during explanations, reacting to content
- "Helps" with tasks (hands tools, holds things—sometimes correctly)
- Gets into things he shouldn't be in
- Occasionally does something genuinely useful, undermining his reputation
- Creates running bits that build over multiple videos

What ALGERNON Doesn't Do

- Talk (no voice, no text-to-speech—expression only)
- Act purely incompetent (he's not a prop, he's a character)
- Dominate videos (he's a presence, not the focus)

Technical Specifications

Joint	Actuation
Base (yaw)	NEMA17 stepper, direct drive
Shoulder (pitch)	NEMA17 stepper, direct drive
Elbow (pitch)	NEMA17 stepper, rolling joint with cammed surface + linkages
Wrist (pitch)	Hobby servo
Wrist (roll)	Hobby servo
End effector	TBD — gripper or tool mount + camera

Component	Details
Controller (Phase 1)	SKR Mini E3 V3 or similar, Klipper/Marlin
Controller (Phase 2+)	Raspberry Pi for high-level control, CV processing
End effector camera	TBD — enables POV shots and future vision-based autonomy

The rolling joint elbow uses a cammed surface with linkages maintaining contact between the cams—a novel mechanism that's technical content in its own right.

Rollout Strategy

ALGERNON debuts in **shortform content first** (Reels, TikTok, Shorts) before appearing in long-form videos. This builds anticipation and establishes the character with an audience before the channel properly launches.

Phase	Content
Pre-launch	Shortform clips: ALGERNON being built, first movements, early mischief
Video 1 (Reducer, Apr 2026)	Workshop presence only — long-form focus stays on the reducer

Video 2 (May 2026)	Dedicated character debut — ALGERNON's own long-form introduction
Ongoing	Recurring presence in workshop content from Video 3 onward

Autonomy Roadmap

Section removed in v2.0 — the autonomy direction for ALGERNON has shifted since v1.4 and needs re-planning before it earns space in the brand guide. For Phase 1 of the channel, assume manual puppeteering for filming; autonomy is developed as a separate thread, not a prescribed three-phase ladder.

Content Strategy

Content Pillars

Type	Description	Frequency
Tentpole Projects	Major builds: actuators, arms, rovers. Full story arc with companion blog.	Monthly
Interstitial	Printer retrofits, tools, "what's on my bench", MSc adjacent.	Bi-weekly
Process/Pivot	"Why I threw out X months of work" — design pivots, failures.	As needed

Companion Content

- **Blog (tomleh.gg)**: Deep technical writeups, equations, design rationale
- **Shortform (self-managed)**: Instagram Reels, TikTok, YouTube Shorts — lighter between tentpoles, ramps up 1-2 weeks before each video launches

Recurring Elements

- **ALGERNON**: 5-DoF workshop arm — see dedicated section above
- **Project status overlays**: Mission control style progress indicators
- **"Kno." end card**: Consistent outro with links to blog/next video

Content Roadmap

Phase 0: Pre-Launch (Dec 2025 - Apr 2026)

Shortform content establishing ALGERNON before long-form launch. Self-managed posting to Instagram Reels, TikTok, YouTube Shorts. Pre-launch extended through Q1 2026 as the original Jan 2026 launch slipped while remaining uni assignments (RASID, CMNT, Robo-ethics, ME52078) took precedence.

Content Ideas:

- ALGERNON build progress clips (printing, assembly, wiring)
- First movements, calibration footage
- Early mischief moments (reaching for things, "investigating" the desk)
- Workshop ambiance with ALGERNON in frame
- Teaser clips with no context — generate curiosity

Platform Strategy:

- TikTok: Primary discovery platform, trending sounds where appropriate
- Instagram Reels: Cross-post from TikTok, build follower base
- YouTube Shorts: Seed the YouTube algorithm, funnel to main channel

Shortform Guidelines:
- Self-managed — prioritise consistency over volume
- Lighter cadence between tentpoles; ramp 1-2 weeks before each video launches
- Maintain visual consistency (NASA-punk aesthetic where possible)
- No voiceover needed — music/ambient audio
- Include watermark on all content (current: "Kno.", under review)

Success Metrics:
- Not about viral hits — about consistent posting and building a small base
- Target: 10-20 shorts posted before Video 1 drops
- Any follower count >0 means Video 1 has an audience

Goal: Build anticipation so Video 1 doesn't drop into a void.

Phase 1: Launch (Apr-Jun 2026)

First three tentpole videos establishing the channel's identity and technical credibility. Monthly cadence.

#	Video	Target
1	Plate harmonic reducer — throwing out months of cycloidal work over a random tweet	24 Apr 2026
2	Meet ALGERNON — dedicated long-form introduction to the workshop arm	End May 2026
3	Rover team content (TBC — likely arm testing or reducer integration)	End Jun 2026

Video 1 — Actuator Pivot (Reducer):
- Hook: "I saw a tweet and threw out months of work"
- Structure: Cycloidal backstory → tweet/paper discovery → why plate harmonic is better → prototype + testing
- ALGERNON: Workshop presence only — shortform has established him; long-form focus stays on the reducer
- Companion blog: Deep dive on reducer theory, paper breakdown
- Target length: 12-18 minutes

Video 2 — Meet ALGERNON:
- Hook: Dedicated introduction to ALGERNON — the workshop arm already glimpsed in shortform
- Structure: Build recap → joint-by-joint tour (including rolling-joint elbow) → first task attempts (helpful + mischievous) → personality reveal → teaser for future videos
- ALGERNON: **The** focus. Character-first framing over technical-first
- Companion blog: Build details, joint design, firmware, control approach
- Target length: 12-18 minutes

Video 3 — Rover Team Content (TBC):
- Likely angle: Rover team arm testing or reducer integration — concept locked mid-quarter based on what ships
- Hook: TBD once concept locked (subject matter may itself be the hook — real competition hardware)

- Structure: Context on rover team involvement → the specific system/subsystem → testing/results → what's next
- ALGERNON: Workshop presence
- Companion blog: Technical writeup of featured system
- Target length: 15-20 minutes
- **Personal NEIL rover remains back-burner — not the Video 3 subject.**

Phase 1 Goals:
- Establish technical credibility
- Establish tone (dry British, personality-driven)
- Introduce ALGERNON as recurring presence
- Build companion blog habit
- Learn the production workflow (filming, editing, publishing cadence)
- **Stretch:** rougher bi-weekly build logs / project updates on Patreon — Joe Barnard-style cadence, gated on channel traction (2k subs as a loose threshold), not a launch requirement

Phase 2: Growth (Jul–Sep 2026)

Assuming Phase 1 completes successfully, expand content variety and posting frequency.

Tentpole Videos (Monthly):
- Continue major project coverage (rover progress, new builds)
- Each tentpole gets companion blog post

Interstitial Content (Bi-weekly):
- "What's on my bench" — 5-8 minute updates on current work
- Tool/technique spotlights — specific skills or tools you're using
- 3D printer retrofit episodes — each printer gets 1-2 videos
- MSc content — research explainers, paper breakdowns

Shortform:
- Continue self-managed shorts from tentpole footage
- ALGERNON clips as ongoing content
- Behind-the-scenes moments
- Ramp around tentpole launches

Phase 2 Goals:
- Increase upload frequency (tentpole + interstitial = 3-4 videos/month)
- Grow subscriber base through consistency
- Identify what content performs best
- Refine production efficiency

Phase 3: Expansion (Oct 2026+)

Longer-term ambitions, contingent on Phase 1-2 success.

Potential Directions:
- Collaboration with other maker channels
- Rover team content (if permission secured)
- SCATT system project (new audience crossover)
- Dissertation algorithm series (multi-part technical deep dive)
- "Making ALGERNON Smarter" premium content
- Sponsorship integration (if audience warrants)

Platform Expansion:
- Evaluate Patreon/membership for premium content
- Consider long-form podcast/discussion format
- Potential conference talks or maker faire presence

Phase 3 Goals:
- Sustainable content creation rhythm
- Revenue generation (sponsorship, membership, or other)
- Recognised presence in robotics/maker community

Gap Content Ideas

Quick-turnaround content for between tentpole videos:

Printer Retrofit Series:
- "Rescuing a dead Ender 3" — diagnosis, repair, upgrades
- Specific upgrade videos (direct drive conversion, Klipper setup, etc.)
- Each printer = 2-3 videos of content

Workshop/Process:
- "What's on my bench this week" — casual updates
- Tool reviews/spotlights (only tools you actually use)
- "How I design [X]" — CAD workflow, design decisions
- Failed prints/parts — what went wrong, what I learned

Technical Explainers:
- MSc coursework adjacent — explain concepts you're learning
- Paper breakdowns — interesting papers in accessible format
- "How [mechanism] actually works" — animations, demos

ALGERNON-specific:
- Calibration and tuning sessions
- New behaviour additions
- "ALGERNON tries to [task]" — task attempt videos

Future Arcs

Multi-video series for long-term planning:

Dissertation Algorithm Series:
- Part 1: The problem (multi-robot coordination)
- Part 2: The algorithm (auction-based task allocation)
- Part 3: Simulation results
- Part 4: Real-world implementation on NEIL
- Potential for 4-6 videos over 6+ months

Novel Cycloidal Design:
- Revisit shelved actuator concept
- "The idea I couldn't let go of"
- Design, prototype, test arc

Rover Team Content (pending permission):
- Behind-the-scenes of competition prep
- Technical deep dives on team systems
- Competition coverage

SCATT Alternative:
- "Building a £50 shooting trainer"
- Appeals to entirely different audience (shooting sports)
- Crossover potential — brings new viewers to robotics content

Production Setup

Hardware

Component	Current Setup
Primary Camera	Panasonic Lumix GH4 + 14–42mm f/3.5–5.6 kit lens
B-Roll / Alt Camera	Nothing 2a Plus (smartphone)
Voiceover Audio	Blue Yeti Nano (desk setup)
On-Location Audio	DJI Mic Mini (2× TX + RX, camera-mounted)

Planned Upgrades

- **Near-term**: Faster prime lens (25mm f/1.7) for low-light workshop shots

Workflow

1. **Filming**: GH4 for main footage, phone for spontaneous B-roll
2. **Audio**: Voiceover-heavy edit using Yeti Nano; accept rough on-location audio
3. **Talking head**: Film at desk (Yeti range) or on location (coastal/garage)
4. **Edit**: Narration-first workflow — write tighter after seeing footage
5. **Review**: Send rough cut to trusted friends for honest feedback

Business Structure

Entity Purpose

- Equipment/expense management
- Collaborator agreements (shortform content creators)
- Future: potential sponsorship handling

Scope Limitation

Corporate structure exists for practical management only. Priority remains: **make videos first, worry about business later.**

IP Considerations

- Plate harmonic reducer design based on patented Korean paper
- US/Korea patents exist; no known EU/UK filing
- Personal/educational use: acceptable
- Closed-source component approach for own projects
- Not open-sourcing reducer design; modular component excluded from public rover project